The four displayed symplectic generators

Human Compression project: L031 corrected recovery

October 2026

We verify the four matrices printed after Theorem 6.1 of Brooksbank–Ferrara–Leemans, in the supplied twelve-page edition of arXiv:1709.02219v2. They do give a rank-four string C-marking for every stated parameter. Their first three generators preserve a *plus*-type quadratic form; thus the preceding minus-type identification is not the identification of this displayed family. The proof below retains the complete ambient support and three-space lemmas of the companion *Endpoint variation in orthogonal symmetry markings*. It does not retain a finite experiment or an orthogonal maximal-subgroup assertion.

1 Parameters, form and two dihedral groups

Let $F = \mathbf{F}_q$, $q \geq 4$ even, and choose $\alpha \in F^\times$ such that

$$C_\alpha = \begin{pmatrix} 0 & 1 \\ 1 & \alpha^2 \end{pmatrix} \quad \text{has order } q+1.$$

Such parameters exist by the norm-one torus construction in the companion. In particular $\alpha \neq 1$, since C_1 has order three. Vectors are rows in the ordered basis (u, e, b, w). Define

$$r_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \alpha & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad r_1 = \begin{pmatrix} 1 & \alpha & \alpha & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1+\alpha & \alpha & 1 \end{pmatrix},$$

$$r_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \alpha \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad r_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \alpha & 1 \end{pmatrix}. \tag{1}$$

Use the quadratic form

$$Q(U, E, B, W) = U^2 + B^2 + W^2 + \alpha UE + \alpha EW + BW. \tag{2}$$

Its polar matrix is

$$\begin{pmatrix} 0 & \alpha & 0 & 0 \\ \alpha & 0 & 0 & \alpha \\ 0 & 0 & 0 & 1 \\ 0 & \alpha & 1 & 0 \end{pmatrix},$$

which is nondegenerate. We write its pairing as $(\ ,\ )$ and $s_v(x) = x + (x, v)v/Q(v)$ for its symmetries. Direct substitution gives $r_0 = s_u$, $r_2 = s_w$, and $r_3(x) = x + \alpha(x, b)b$. Put

$$l = e + b, \quad h = (1+\alpha)\alpha^{-1}e + b, \quad y = e + \alpha b, \qquad L_u = \langle u, l \rangle, \quad L_w = \langle w, h \rangle.$$

The values $Q(u), Q(l), Q(w), Q(h), Q(b)$ are one, while $Q(y) = \alpha^2$. Moreover $(u, l) = (w, h) = \alpha$, and

$$r_1 = s_l s_b = s_h s_y. \tag{3}$$

Both identities follow by applying the two sides to u, e, b, w; their two symmetry factors commute. In particular the first three matrices preserve Q. All four preserve its polar form and are nonidentity involutions. The nonadjacent pairs in (1) commute.

The characteristic polynomial of $s_u s_l$ on L_u, and of $s_w s_h$ on L_w, is $X^2 + \alpha^2 X + 1$. These elements have order $q+1$, by the condition on C_α. Hence L_u, L_w are asingular and their displayed symmetry pairs generate their whole dihedral isometry groups. Since $b \perp L_u$ and $y \perp L_w$, (3) and the oddness of $q+1$ give

$$\langle r_0, r_1 \rangle = O(L_u) \times \langle s_b \rangle, \qquad \langle r_1, r_2 \rangle = O(L_w) \times \langle s_y \rangle. \tag{4}$$

These are actual ambient internal direct products, of order $4(q+1)$: for example $(r_0 r_1)^{q+1} = s_b$. Both factors fix the perpendicular complement of their stated support.

The type of (2) is explicitly plus. Indeed $v = w + (1 + \alpha^{-1})u$ lies in $L_u^\perp$ and

$$Q(v) = \alpha^{-2}, \quad Q(b) = 1, \quad (v, b) = 1.$$

The two perpendicular planes L_u and $\langle b, v \rangle$ have the same Arf trace $\mathrm{Tr}(\alpha^{-2}) = 1$. Their sum therefore has Arf trace zero. In particular the distinction from the source's minus-type discussion is a uniform calculation, not an inference from a sampled order.

2 Full generation

Let $H = \langle r_0, r_1, r_2 \rangle$. Equation (4) supplies Σ_{L_u} and s_b. The space $T = \langle u, l, w \rangle$ is orthogonal of dimension three. Its radical is generated by $(\alpha + 1)u + \alpha w$, of quadratic value one. Also w is nonradical in T, since $(w, l) = \alpha + 1 \neq 0$. The companion's ambient three-space lemma gives $\Sigma_T \leq H$.

The asingular plane $L_u^{s_w}$ contains $l + (\alpha+1)w$, whose pairing with b is $\alpha + 1 \neq 0$. Consequently its conjugate by s_b is an asingular plane not contained in T: the changed vector acquires the nonzero b-component $(\alpha + 1)b$ modulo T. The companion's hyperplane saturation lemma now gives

$$H = O(V, Q) = O^+(4, q). \tag{5}$$

To see that $G = \langle H, r_3 \rangle$ is the entire symplectic group without using maximality, choose $t \neq 0$ with $t^2 = (\alpha^2 + \alpha)^{-1}$ and set

$$a = (0, (1 + t^2)/(\alpha t), 0, t).$$

Then $Q(a) = 1$, $(a, b) = t$, and

$$Q(a r_3) = Q(a) + (\alpha^2 + \alpha)(a, b)^2 = 0.$$

Thus conjugating $s_a \in H$ by r_3 gives a transvection with a nonzero singular axis. The full plus orthogonal group is transitive on its nonzero singular vectors: in the hyperbolic model $Q(X) = \det X$ on 2 by 2 matrices, these are the nonzero rank-one matrices, and left and right $\mathrm{SL}_2(q)$ actions are transitive on them. Scaling an axis scales its transvection parameter by a square; all nonzero parameters therefore occur on every singular axis.

These transvections generate every transvection. For a nonsingular z, choose a singular x with $(z, x) \neq 0$; singular vectors span the hyperbolic space, so this is possible. The vector $z' = z + Q(z)x/(z, x)$ is singular. A transvection on x sends z' to z. Conjugation consequently supplies every parameter on z. The elementary transvection generation in the companion proves

$$G = \mathrm{Sp}(4, q). \tag{6}$$

3 All intersections

First consider the three marks of H. Put $U = \langle u, e, b\rangle$ and $W = \langle w, e, b\rangle$. Elements of the two groups in (4) have support in U and W respectively; their intersection is supported in $\langle e, b\rangle$. An element of the first group supported there preserves L_u and has its image on L_u contained in $\langle l\rangle$. The one-space support assertion in the companion makes its action on L_u either 1 or s_l. Its complementary factor is 1 or s_b. Thus it belongs to $\langle s_l, s_b\rangle$. Likewise the second group contributes only $\langle s_h, s_y\rangle$. The root lines $\langle l\rangle, \langle b\rangle, \langle h\rangle, \langle y\rangle$ are pairwise distinct when $\alpha \notin \{0, 1\}$. The first two Klein four groups have, by (3), their sole common nonidentity element r_1. Their intersection is therefore exactly $\langle r_1\rangle$. This proves that (r_0, r_1, r_2) is C.

For the other end group $K = \langle r_1, r_2, r_3\rangle$, the second group in (4) preserves L_w and its perpendicular complement. If one of its elements is also in $\langle r_2, r_3\rangle$, its support lies in $\langle w, b\rangle$. On L_w it is supported on $L_w \cap \langle w, b\rangle = \langle w\rangle$, so is 1 or s_w there. Its possible other factor is s_y, whose root is outside $\langle w, b\rangle$. Hence the intersection is exactly $\langle r_2\rangle$. Both adjacent two-generator groups consist of distinct involutions, so this proves that (r_1, r_2, r_3) is C.

It remains to intersect H and K. Every element of K is supported in W, and preserves on W the quadratic form

$$Q'(E, B, W) = \frac{\alpha}{\alpha + 1}E^2 + \alpha^{-1}B^2 + W^2 + \alpha EW + BW. \tag{7}$$

This is checked directly on its three generators. The difference $Q'|_W - Q|_W$ is a nonzero squared linear functional whose kernel is L_w; explicitly it vanishes on w, h and is nonzero on y. Thus an element of $H \cap K$ preserves L_w and Q, and has support in W. On the perpendicular decomposition $V = L_w \perp L_w^{\perp}$, its first action belongs to $O(L_w)$. Its second is supported in $W \cap L_w^{\perp} = \langle y\rangle$, and is therefore 1 or s_y. By (4) this yields

$$H \cap K = \langle r_1, r_2\rangle. \tag{8}$$

The standard string end-parabolic criterion proves the full intersection property for (1).

Theorem 1. *For every even $q \geq 4$ and every source parameter α with $|C_\alpha| = q + 1$, the four literal matrices (1) form a rank-four string C-marking of* $\mathrm{Sp}(4, q)$. *Their first-three subgroup is $O^+(4, q)$, of order $2q^2(q^2 - 1)^2$, rather than the minus group of order $2q^2(q^4 - 1)$.*

The marking is irreducible: a commuting adjacent pair in a string C-marking would split the generated group into two nontrivial direct factors, whereas $\mathrm{Sp}(4, q)$ is simple for $q \geq 4$. Alternatively noncommutation is immediate in (1). The source's principal existence conclusion is preserved. This calculation does not assert that every preliminary geometric choice in its preceding paragraph has the same type, and does not claim an historical new construction.